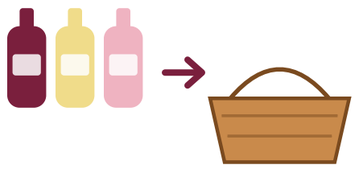
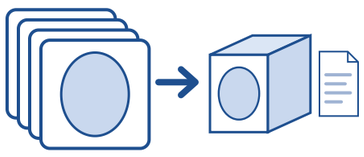
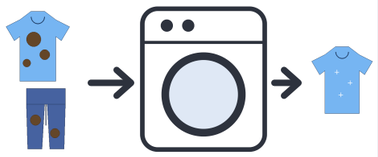
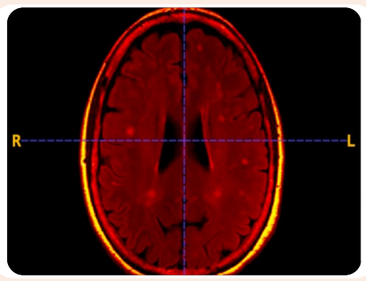
WHY GAMES

Learning Real Lab Skills Through Play

The early games use everyday objects; the later games use real MRI data

IN THE GAME		IN A NEUROIMAGING LAB
	LESSON 1 · VIRTUAL HOUSE **Furniture files in rooms** Move objects into the right rooms, then find them again.	**A study folder tree** “MRI projects are folder-heavy. Before students can run analysis software, they need to know where they are, what files are present, and how to move directly to the right place.” — House Game FAQ `raw_dicoms/sub-PD01/FLAIR_DICOM`
	LESSON 2 · WINE SHED **Wines that match an order** Read each bottle’s color and flavor; move the matches into a customer’s basket.	**Selecting scans that meet criteria** “MRI labs constantly search through folders, identify files that match criteria, and process them consistently.” — Wine Game FAQ `for SUBJECT in $SUBJECTS; do`
	LESSON 3 · SCAN CONVERTER **Real scans, no metaphor** Convert one participant’s raw DICOM folders into NIfTI images and review them.	**The first real step of MRI analysis** “This game teaches one of the first real transformations in MRI analysis: DICOM folder → NIfTI image + JSON metadata.” — Scan Converter README `dcm2niix -z y -b y -f "${SUBJECT}_FLAIR" …`
	LESSON 4 · VIRTUAL LAUNDROMAT **Dirty in, clean out** Wash one load by hand, then many with wildcards and loops, then submit a job to a simulated remote laundromat.	**A processing pipeline on a cluster** “Raw data goes in, a script is processing the data and the outputs are produced and then we shortly verify those results.” — Osiel, Laundry FAQ video, 00:34 `ssh → scp → sbatch → read the log`
	LESSON 5 · BRAIN IMAGE ANALYSIS **Six participants, two groups** Convert six participants, mark lesion-like spots on FLAIR, measure and compare groups.	**A small group analysis** “We are not diagnosing lesions. We are practicing image inspection, manual annotation, measurement, and group comparison.” — Lesion game README `results/group_comparison.txt`